

# Native VLAN

## Project Overview

This project was initiated to explore the functionality and significance of the **Native VLAN** in Ethernet networks utilizing IEEE 802.1Q trunking. The main goal is to understand how the Native VLAN manages untagged traffic on trunk links and to identify best practices and potential security risks associated with its configuration. Key objectives include:

- Understand the role of the Native VLAN as the default VLAN for handling untagged frames on trunk ports.
- Review how IEEE 802.1Q tagging differentiates between tagged VLAN frames and untagged Native VLAN frames during trunk transmission.
- Examine step-by-step configuration of Native VLANs on Cisco switches, including enabling trunking, assigning the native VLAN, and verifying configurations.
- Analyze the impact of Native VLAN mismatches, including misdirected traffic and VLAN hopping security risks.
- Review the use of allowed VLAN lists on trunk ports to restrict VLAN traffic over trunk links.
- Identify affected network locations such as trunk ports, access ports, inter-switch links, Layer 3 routing interfaces, and protocol traffic (e.g., CDP, STP, VTP).
- Apply best practices, including avoiding VLAN 1 as the native VLAN, using unused VLAN IDs, explicitly configuring native VLAN on both trunk ends, disabling unused ports, employing VLAN pruning, implementing VLAN access control lists (VACLs), and regularly monitoring VLAN and trunk configurations.

## Description

A Native VLAN is the default VLAN that handles untagged traffic on trunk links in a network. Trunk links are simply connections between switches or between a switch and a router. These links let multiple VLANs share a single network connection by tagging Ethernet frames using the IEEE 802.1Q standard. The Native VLAN acts as a "home" for untagged frames—data packets without a VLAN ID. Instead of being discarded, these untagged frames are assigned to the Native VLAN, allowing them to be processed correctly.

## Recommendations

- Avoid using VLAN 1 as native VLAN
- Use an unused VLAN ID for native VLAN
- Explicitly configure native VLAN on both ends of a trunk link
- Disable unused switch ports or assign them to an unused VLAN
- Use VLAN pruning and VTP pruning if available
- Implement VLAN access control lists (VACLs)
- Regularly monitor VLAN configurations and trunk ports

## References

<https://www.ruijie.com/en-global/support/tech-gallery/what-is-native-vlan>

<https://jumpcloud.com/it-index/what-is-native-vlan>

<https://www.networkacademy.io/ccna/ethernet/trunk-native-vlan>

<https://www.geeksforgeeks.org/computer-networks/native-vlan/>

## Details

If a switch receives a frame without any tag, it automatically assumes that it belongs to the native VLAN. Trunk links exclusively transmit tagged frames, enabling the transfer of data between different VLANs. By using the ping command, network administrators can observe this behavior. A trunk can only have one native VLAN, and this VLAN must match on both ends of the trunk. The native VLAN is responsible for carrying all untagged frames, earning it the moniker of the "untagged VLAN."

The Native VLAN is one conception defined in the 802.1 q grade that was created for backward compatibility with the old device that does not endorse VLANs. The concept at the back of the local VLAN is that a few network traffic still flows over the entire switched network even if the trunk fails.

The Native VLAN is a VLAN number configured per trunk port; it is locally significant, and it instructs the switch the following:

*"If you receive an untagged frame on this trunk port, forward it as if it is part of the native VLAN number."*

Recall that a trunk port sends and receives Ethernet frames tagged with IEEE 802.1q VLAN tags. The primary idea behind this is to be able to transport frames from multiple VLANs over a single physical link between switches. This means that both ends of a trunk will always receive tagged frames, as shown in the diagram below.

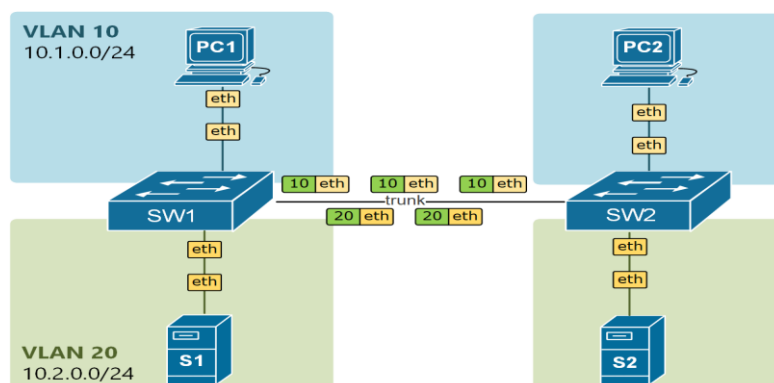


Figure 1. Frames forwarding over a trunk line. (animated)

## *Native VLAN and 802.1Q Tagging*

IEEE 802.1Q tags Ethernet frames with VLAN IDs for proper routing. However, Native VLAN traffic is sent untagged across trunk links.

- Native VLAN frames: Sent untagged on trunk ports.
- Tagged VLAN frames: Include VLAN ID in the 802.1Q tag.
- At the receiving switch: Untagged frames are treated as Native VLAN traffic.

## *Setting Up Native VLAN*

1. Access Interface:

```
Switch# configure terminal  
Switch(config)# interface gigabitethernet 0/1
```

2. Enable Trunking:

```
Switch(config-if)# switchport mode trunk
```

3. Set Native VLAN:

```
Switch(config-if)# switchport trunk native vlan [VLAN_ID]
```

4. Verify:

```
Switch# show running-config | include interface
```

## *Native VLAN Mismatch & Allowed VLANs*

A Native VLAN mismatch happens when trunk ports on both sides use different native VLANs (e.g., VLAN 10 on one side, VLAN 20 on the other). This leads to:

- Misdirected traffic – Untagged frames may be assigned to the wrong VLAN.
- VLAN hopping – Potential security vulnerability.

If CDP is disabled, switches can't detect the mismatch.

## *Allowed VLANs on a Trunk*

By default, trunks carry all VLANs, but you can restrict them:

```
Switch(config-if)# switchport trunk allowed vlan 10,20
```

This ensures only specific VLAN traffic (e.g., 10 and 20) is passed across the trunk.

## **Affected Locations**

- Trunk Ports
- Access Ports Connected to End Devices
- Inter-Switch Links
- Router or Layer 3 Switch Interfaces
- Protocol Traffic (CDP, STP, VTP)